Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6				**Indicative content** **Boiling** • boiling causes calcium hydrogencarbonate to decompose (or magnesium hydrogencarbonate) • a precipitate of calcium carbonate is formed (limescale) • calcium / magnesium ions are removed from the water • $Ca(HCO_3)_2 \rightarrow CaCO_3 + H_2O + CO_2$ • very simple method of softening • but limescale is formed which causes 'furring' / reduces boiler efficiency / blocks pipes • boiling only removes temporary hardness • not for large-scale as it requires a lot of energy references to 'scum' are neutral **Ion exchange** • ion exchange columns contain a resin (in small balls) coated with sodium ions • as hard water passes through the column, sodium ions come off the resin and go into the water, while calcium / magnesium ions come out of the water and stick to the resin • removes both temporary and permanent hardness • continuous process • resin needs changing / regeneration • water contains high levels of sodium ions which can be harmful for some people	6			6		3